



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 3	
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Batch	2023 - 2027
Class	CSE – B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Simulating Git Collaboration & Conflict Resolution
Implementation & Theory	<pre>• PS E:\DevOps_SEM_7> git checkout main Switched to branch 'main' Your branch is up to date with 'origin/main'. • PS E:\DevOps_SEM_7> git branch feature/user-auth * main PS E:\DevOps_SEM_7></pre> Switched to the main branch to simulate concurrent development. This allows different changes to be made on separate branches, creating a realistic merge conflict scenario. <pre>app.py 1, M X Assignment - 2 > app.py > ... 1 from flask import Flask 2 3 app = Flask(__name__) 4 5 @app.route("/") 6 def home(): 7 print("Main Branch Update") 8 return "Hello World!" 9 10 if __name__ == "__main__": 11 app.run(debug=True)</pre>

Modified the same line of code on the **main** branch that had previously been changed on the feature branch. This creates a conflicting change that Git cannot merge automatically, requiring manual conflict resolution.

```
PS E:\DevOps_SEM_7> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Assignment - 2/app.py

no changes added to commit (use "git add" and/or "git commit -a")
```

Verified the repository status using git status. Git detected the modified file, confirming that the changes are ready to be staged and committed.

```
PS E:\DevOps_SEM_7> git add .
>> git commit -m "Updated app on main branch"
[main e50beb1] Updated app on main branch
 1 file changed, 1 insertion(+)
PS E:\DevOps_SEM_7>
```

Staged and committed the changes made to the main branch. This creates a separate commit history that differs from the feature branch, setting up the conditions for a merge conflict.

```
Auto-merging Assignment - 2/app.py
CONFLICT (add/add): Merge conflict in Assignment - 2/app.py
Automatic merge failed; fix conflicts and then commit the result.
```

Attempted to merge the feature/user-auth branch into main. Since both branches modified the same section of the file differently, Git detected a merge conflict and requested manual resolution.

```

app.py 1 X
Assignment - 2 > app.py > ...
1  from flask import Flask
2
3  app = Flask(__name__)
4
5  @app.route("/")
6  def home():
7      print("Main Branch Update")
8      print("User Authentication Feature Added")
9      return "Hello World!"
10
11 if __name__ == "__main__":
12     app.run(debug=True)

```

Manually resolved the merge conflict by editing the conflicting section and preserving the required changes. After removing the conflict markers, the file became a valid Python program.

```

PS E:\DevOps_SEM_7> git add "Assignment - 2/app.py"
PS E:\DevOps_SEM_7> git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

All conflicts fixed but you are still merging.
(use "git commit" to conclude merge)

```

Staged the resolved files to mark the merge conflict as resolved. Git now recognizes that all conflicts have been handled successfully.

Verified that all merge conflicts had been resolved successfully. Git confirmed that only the merge commit remained to complete the integration.

```

PS E:\DevOps_SEM_7> git commit -m "Resolved merge conflict between main and feature/user-auth"
[main b19ef6d] Resolved merge conflict between main and feature/user-auth

```

```

PS E:\DevOps_SEM_7> git push origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 796 bytes | 796.00 KiB/s, done.
Total 8 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 3 local objects.
To https://github.com/AryanSheladia/DevOps_SEM_7
055dfd2..b19ef6d  main -> main

```

Created the merge commit to finalize the integration of the feature branch into the main branch. This commit records the manual conflict resolution in the repository history.

Pushed the merged changes to the remote GitHub repository, ensuring the local and remote repositories are synchronized with the latest updates.

```
055af4277b19c76d main -> main
PS E:\DevOps_SEM_7> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Verified that the repository is in a clean state after the merge and push operations. This confirms that all changes have been committed and synchronized successfully.